

The Bond Market

16.1 INTRODUCTION

Bonds are tradeable loans issued by governments, corporations, and financial institutions to raise long-term funds. Strictly speaking, there is no important legal difference between money market paper and bonds but the periodic interest paid on bonds makes the valuation slightly more complicated. While issuers are in principle free to design bonds in any way they wish, it is in practice much easier to raise debt when it has a form that is already known to potential investors. This is why bonds tend to have certain common cashflow characteristics and why standard terms reappear in the legal documentation of bonds.

Bonds establish payment and other obligations for the issuer, as well as certain actions the issuer can take when interacting with bond holders. The total set of rights and obligations of issuer and bond holder derives from multiple sources. The most important is the governing law of the jurisdiction where the bond is issued because every jurisdiction has a set of laws dealing with debts. For sovereign issuers in their own jurisdiction, a combination of civil code and specific debt issuance laws usually complete the legal conditions of each bond. It should be noted that sovereigns can change the governing law which means that in principle they can change the terms of existing debt. Although this option can be used, and has been used, to reduce debt obligations, sovereigns using this option typically lose access to capital markets for some time.

Private issuers, or sovereign issuers issuing outside their own jurisdiction, generally need to provide more specific legal terms for bonds which is generally done in a document called the prospectus. A prospectus combines information about the issuer with specific legal terms and conditions of one or more bond issues. Typical clauses that can appear in a prospectus are discussed further below. A prospectus is a legal document and the contents of this document are subject to strict regulation in many jurisdictions. Such regulation generally aims at establishing a common minimal set of information disclosure about the issuer rather than specific legal provisions for the bonds themselves. The production of a prospectus is expensive because it requires a lot of legal and accounting work. Issuers can save costs by producing only one prospectus for a series of bonds that they plan to issue. Such a series of bonds is called a programme and correspondingly the prospectus would be called a programme or base prospectus. Depending on the type of debt issued under a programme, the programme itself might be referred to as for instance an EMTN (euro medium-term note) or CP (commercial paper) programme.

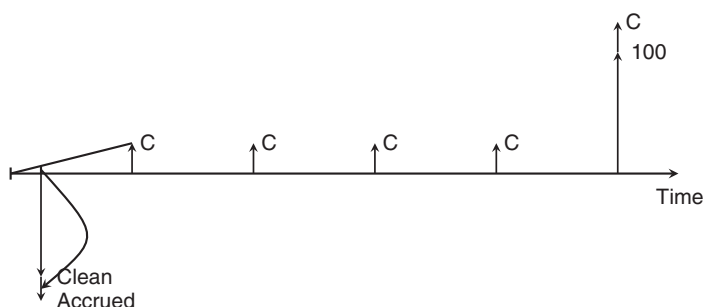


FIGURE 16.1 Simple annual fixed rate bullet bond structure. The bond repays its par amount at the maturity date. The bond has regular accrual periods so every coupon payment is equal to the nominal coupon C . A purchase transaction is indicated with negative cash flows (the consideration paid to the seller of the bond) and the consideration is split into clean price and accrued interest.

When issuing under a programme, issuers issue new bonds with simplified terms and conditions documents that simply reference the relevant programme prospectus.

Bonds can be classified along a number of criteria, such as their cashflow structure, issuer type, or legal features. From an economic point of view, the cashflow structure is the most important feature of a bond because it determines the valuation approach used to price and hedge the security. For visualisation purposes, it is convenient to think of the cashflows as arrows on a timeline as shown in Figure 16.1.

It should always be borne in mind that while cashflows are fungible for the purpose of valuing bonds in the absence of default risk, there is an important legal difference between coupon and redemption payments in defaults. In most jurisdictions insolvency triggers an immediate claim on payment of the remaining principal due to investors, plus any accrued interest. These claims exclude the right to payment of future coupons that have not yet started accruing. The present value of future coupons therefore usually declines when the credit quality of a bond deteriorates because it becomes less likely that they will ever get paid. The present value of the principal might instead even increase because it will get paid earlier (albeit probably only in part).

16.2 CASHFLOW TYPES

16.2.1 Bullet bonds

Bonds usually pay interest in the form of coupons at regular intervals called coupon periods. Depending on the length of coupon period, bonds are classified as annual, semi-annual, quarterly, etc. Coupons are always stated in annualised terms, so a semi-annual bond will pay half the stated coupon twice a year, a quarterly bond will pay a quarter of the coupon four times a year and so on. The distribution of coupons to investors creates an administrative burden which makes more frequent coupons more expensive to manage. On the other hand, both issuers and investors tend to find it inconvenient to deal with few large cashflows per year. This would favour higher

coupon frequencies. The compromise eventually chosen by issuers reflects the relative importance of these concerns. Note that institutional bond portfolios (and the liabilities of larger issuers) can avoid bunching of cashflows by diversifying across many bonds.

Coupon rates, expressed in percentages, are in most markets binary, rather than decimal, fractions for historical reasons. Usually, these fractions are multiples of eighths because bonds are often issued in certificates referring to 1,000 in the local currency (euros, dollars, etc), and the smallest fraction that can be paid in whole cents (i.e., the local version of the small denomination) is one eighth¹. Binary coupons have the added advantage of some redundancy when expressed in decimal form. For instance, a coupon of 3.152 on a bond from an issuer that usually has binary fractions can usually be identified as a typo. Decimal coupons are used for government bonds in Japan, Spain, and other countries, and private issuers sometimes choose decimal fractions down to 1/1,000th of a percent to place new bonds exactly at par value.

Even with fixed rate bullet bonds, cashflow structures can have minor complications. It is quite common that the time period between the first accrual day (the day when a bond begins to accrue an interest payment obligation for the issuer) is longer or shorter than a regular coupon period. In such cases of so-called long first coupon or short first coupon bonds, the first coupon payment is adjusted upwards or downwards, respectively, to reflect the different accrual period. Less commonly, the last coupon period can be shorter than a regular coupon period, with a correspondingly smaller last coupon period. Such structures occur for instance in Japanese municipal bonds.

A related class of bonds, floating-rate notes, have regular coupon payments that are variable. These FRNs are discussed in Chapter 17 because their valuations are very different from fixed coupon bonds. Inflation-linked bonds, where in most cases a fixed coupon is adjusted for changes in some price level, also merit a separate chapter, Chapter 23.

16.2.2 Zero coupon bonds, perpetuals and annuities

A special class of fixed rate bond pays no interest at all and is known as a zero-coupon bond, or zero. The equivalent of interest income for these bonds arises because they are issued below par, generating a capital gain towards redemption.



Aside from money market instruments, zeros tend to be created as the result of stripping coupon bonds, i.e., splitting coupon bonds into separate payments of coupons

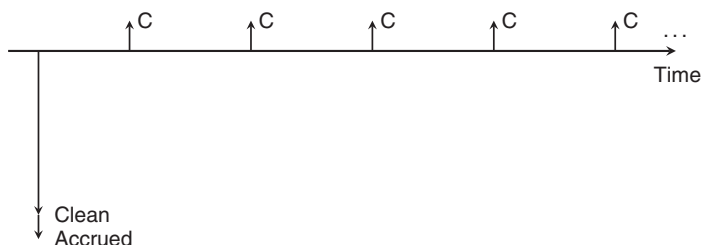
¹A sixteenth of a percent of 1,000 euros is 62.5 cents.

and principal². It should be borne in mind that a zero coupon bond is different from a coupon bond with a fixed coupon of 0% even though both only have a single payment flow at maturity. While a zero has no natural coupon frequency, a 0% fixed rate bond has the same coupon frequency as a related bond with a coupon above 0%. For instance, a 0% BTPS (Italian government bond) would naturally have a semi-annual coupon frequency while a 0% OBL (German government bond) would have an annual coupon frequency even though neither actually pays a coupon. The coupon frequency can affect the calculation of yields and is therefore important. For zero coupon bonds, the coupon frequency tends to be derived from related securities but is not as straightforwardly defined.

Another important aspect of zero coupon bonds is their tax and accounting treatment. Many jurisdictions treat interest income differently from capital gains. Some investors also prefer to have income deferred, for instance because they work in a progressive income tax system and would like to be able to pay tax on their investment income when they are retired and in a lower tax bracket. Zero coupon bonds appear ideal from this point of view because investors realise their return when the bond repays. Tax authorities are of course aware of this issue and accordingly tend to treat zeros differently from coupon bonds.

In accounting terms, a zero coupon bond that is a coupon stripped from a coupon-bearing bond is not treated as a long-term liability of the issuer in the same way that the principal obligation is. Interest payable at market rates is recorded differently from the principal obligation, and creating what looks like a principal zero-coupon payment from a coupon does not change the indebtedness of the issuer.

In some way, the opposite of zero coupon bonds are perpetual bonds, also called perpetuals.



While a zero only makes a principal repayment but pays no coupons, a perpetual bond pays only coupons but never repays the principal. The only way to redeem a perpetual bond is a buy-back by the issuer (here, the holder has to be willing to sell), or, if this is part of the bond prospectus, a call (an early repayment at the sole discretion of the issuer). Sometimes, the word consol is used interchangeably with the word perpetual but this is somewhat inaccurate. A consol, or more explicitly a consolidated loan, is a form of United Kingdom government debt that was created by consolidating various

²The (most certainly apocryphal) acronym STRIPS stands for Separate TRading of Interest and Principal Securities. The first strips were created by physically stripping coupons from paper bond certificates.

earlier outstanding annuities into a single issue in 1754 [57]³. Several consols were issued over the years but the oldest remaining consol, the $2\frac{1}{2}\%$ Annuities, created in 1853, was finally redeemed on 5 July 2015. Other perpetual bonds from private issuers remain, but typically have call features that all but ensure a finite expected lifetime. Perpetuals that could be redeemed at the discretion of the borrower were used in 18th century France as a way of circumventing usury laws because the absence of a fixed repayment date made these instruments look less like a loan [60].

Perpetuals are an example of an annuity, a bond that pays fixed cashflows at regular intervals⁴. While perpetuals have an infinite maturity, other annuities make regular payments that mix principal repayments with interest payments. Because the repayment of principal happens over time, such bonds are also called amortising bonds.

The most common form of a finite maturity annuity is an amortising mortgage where the annuity payments P are calculated according to the formula:

$$P = N_0 \frac{r}{1 - (1 + r)^{-n}} \quad (16.1)$$

for n payments (i.e. for semi-annual payments over N years, $n = 2N$) and an annualised interest rate of $c = fr$ (i.e., $f = 2$ for semiannual payments of $f = 12$ for a usual consumer mortgage). The initial notional of the bond is N_0 . Note that banks tend to use this formula when quoting mortgage repayment amounts to customers even when interest rates are reset periodically so that c is not constant over time as assumed in Equation (16.1).

Based on Equation (16.1), one can calculate the remaining principal amount of the annuity in period k ($0 \leq k \leq n$):

$$N_k = N_0 \left[(1 + r)^k - \frac{(1 + r)^k - 1}{1 - (1 + r)^{-n}} \right] \quad (16.2)$$

with the special values $N_k = N_0$ for $k = 0$ and $N_n = 0$. It should be remembered that the standard convention is to quote bond prices in percentages of the remaining principal outstanding. Here, this would be N_k , not N_0 .

The interest amount I_k that is paid on payment date k is simply the product of r and the principal outstanding at $k - 1$, N_{k-1} . Specifically,

$$I_k = N_0 \frac{r[(1 + r)^n - (1 + r)^{k-1}]}{(1 + r)^n - 1} \quad (16.3)$$

The amortisation payment A_k made at payment date k , $P - I_k$ is:

$$A_k = N_0 \frac{r(1 + r)^{k-1}}{(1 + r)^n - 1} = P - I_k \quad (16.4)$$

³According to the UK Debt Management Office, this 3% annuity was first issued in 1752.

⁴The word annuity does not imply an annual coupon frequency. UK perpetuals were called annuities although they had semi-annual or quarterly coupons.

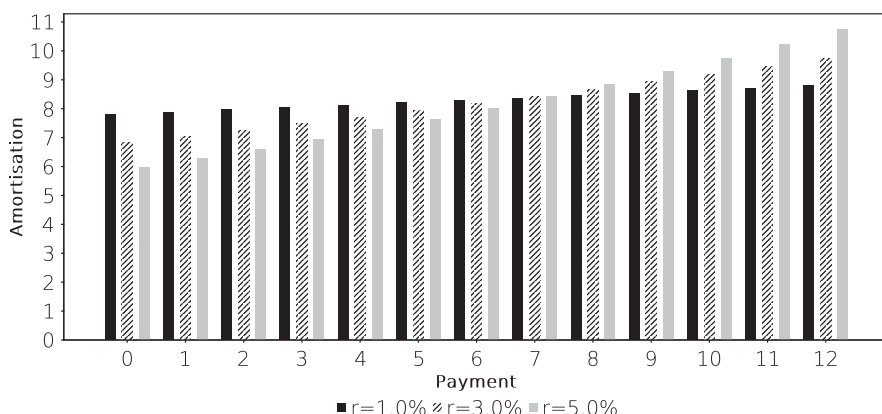


FIGURE 16.2 The amortisation payments A_k of an annuity for different interest rates ($n = 12$). The higher interest rates are the more back-loaded the repayment of principal becomes.

Examples of bonds following this convention are Icelandic Housing Finance Fund (HFF) bonds. These are semi-annual bonds with a quoted interest rate of 3.75%. In this case, $r = 1.875\%$. The payments P depend on the initial maturity of the bonds. For the 2024 bond, which had an initial maturity of 20 years, $n = 40$ so that $P = 3.5759128$ while for the 2034 bond with an initial maturity of 30 years, $n = 60$ and $P = 2.7903945$. This highlights that for finite maturity annuities, the stated interest rate does not provide an immediate clue on the bond cashflows. A 3.75% US Treasury bond would simply pay 1.875% coupons twice a year and par at maturity.

Note that while Equation (16.1) does not allow $r = 0$ directly, one can use the observation that for small r , $1/(1+r) \approx 1-r$ to work out that $\lim_{r \rightarrow 0} P = N_0/n$. Intuitively, for zero interest rates a finite maturity annuity is simply a repayment of the original principal in equal installments. For higher interest rates, the repayment of principal is pushed backwards (cf. Figure 16.2).

In contrast to annuity bonds following the standard amortisation convention, straight-line amortising bonds have cashflows that comprise interest and a fixed share of the original principal being amortised. Cashflows therefore vary (decline) even when the bonds have a fixed interest rate.

16.3 ISSUER TYPES

Bonds can be issued by a variety of legal entities. Typically, governments issue debt under their own domestic law and these bonds form the reference for debt in that domestic market. However, governments can also issue bonds in different jurisdictions where they compete with other issuers, including the government of that jurisdiction.

In many countries, local entities below the central government, such as regions or cities, also access the bond market under their own name. Typically, funding for such entities is a mix of own-name bond issuance in the so-called municipal or sub-sovereign bond market, loans from special funding vehicles that pool municipal

debt and issue in the bond market under their own name, and bank loans of various types. Many countries have government-supported specialist banks that act as funding vehicles for municipalities, such as Dexia Municipal Agency in Belgium, or Agence France Locale in France. For instance, Japanese municipalities (prefectures and large cities) fund through:

- bond issues under their own names,
- joint municipal bond issues in varying compositions,
- loans from the Japanese Funding for Municipalities agency (JFM) which is jointly owned by the municipalities,
- bank loans.

German regional entities (Länder) can raise funds from:

- bond issues under their own names,
- joint Länder issues in varying compositions,
- bank loans.

A single joint issue of the German federal government and the Länder exists but has been judged not to be cost effective. Unlike Japanese municipalities, which are liable jointly for all of the outstanding joint municipal bonds they participate in, German Länder are only liable for their share in each issue which means that each bond issue has a different credit risk structure. In practice, few investors view this as a problem.

In the United States in particular, there is a set of municipal bonds that is tied to specific investment projects, the so-called project bonds. Unlike a standard municipal bond, where investors have the general revenues of the issuing entity as sources of repayment to rely on, project bonds are linked to the revenue of a specific project, such as a toll bridge or sewage system. There may in some instances be some credit support from the issuing municipality, but in all cases investors need to be aware of the specifics of that project and the financing sources. Project bonds have had mixed success in other countries.

Banks form the next larger issuer group and use their bonds to refinance their loan and investment activity. International banks tend to be corporate entities with a large number of subsidiaries, some of which are pure funding vehicles. Investors need to pay attention to the specific legal entity that issues a particular bond, and how that entity is being supported by the overall corporate structure. Various types of support exist, from outright guarantees to so-called letters of comfort which are a weaker form of support. Bank bonds are often held by other banks as a form of investment because bank debt has preferential treatment under some bank capital regulations. A specific form of bank bond, covered bonds, is discussed separately in Chapter 25.

In the US, specialist companies have been set up to source cheap funding for mortgage loans. While these are known in the US as agencies, only some of the issuers are actually government agencies. The two largest issuers, Fannie Mae and Freddie Mac (after their acronyms FHLMC and FRMC) are private-law corporations and therefore more correctly named government-sponsored entities (GSEs). The term agency is in other jurisdictions reserved for issuers that are indeed public sector entities, such as CADES in France or Landesrentenbank in Germany.

Non-financial corporations issue bonds as an alternative to loan funding. As such, only corporations that can successfully appeal to investors have access to the bond market. Such corporations tend to have long-established names, large and stable cash flow, and stable earnings. The largest corporate issuers are therefore utilities, car manufacturers, airlines, etc., while newer technology companies rarely access the bond market. Companies with higher growth rates tend to rely more on bank loans and various classes of equity for funding. As with banks, investors need to be aware that a given corporation may issue bonds from a funding vehicle, a special-purpose company in a tax-friendly jurisdiction⁵. Exactly how the funding vehicle is able to repay bonds from the funds of the parent company is important for investors.

Smaller companies can issue bonds but the market for such debt tends to be less stable. Germany, for instance, has had a relatively vibrant market for bonds issued by medium-sized companies but a spate of high-profile defaults has diminished the standing of this asset class.

In some cases, legal entities are created solely for the purpose of issuing debt that refinances particular assets. Such entities are called special-purpose vehicles, or SPV and the resulting debt instruments are usually called asset-backed securities or ABS. The creation of an SPV can have multiple purposes. The buyer or a particular bond may have a preference for a different cashflow structure and ask an investment bank to repackage that bond with a set of derivatives to change the cashflow structure. In other cases, the SPV is set up to be the legal owner of a set of cashflows and exists to sell these cashflows, after suitable transformation, to investors. In either case, the assets in the SPV, and obligations that the SPV may have besides the debt it issues, are important considerations for investors.

16.3.1 Joint issuance

There are some instances where multiple legal entities issue joint bonds. This means that a single bond provides funding to multiple entities. It also means that multiple entities are responsible for the repayment and coupons of these bonds. Examples of such issues are Länder bonds issued by varying consortia of German states, and joint municipal bonds issued by Japanese prefectures. Beyond such examples of joint issuance by entities of similar properties, there is a single example of a bond issued jointly by some German states and the German federal government.

The proceeds of the bond issuance are shared among the issuing consortia according to some pre-determined key and under normal circumstances that key also determines the relative share of the debt service obligation. When one of the participants of the consortium is no longer able to service the debt, the question arises what happens with the corresponding share of the obligation. There are two fundamental models known as several liability, and joint and several liability. Under several (but not joint) liability, the entities of the consortium are liable for their share of the liability but not those of other members. Under joint and several liability, each member of the consortium is liable for the entire issue. Joint and several liability is a substantial source of

⁵The Netherlands are a common jurisdiction for many corporate issuers.

risk for consortium members because they may have to repay the obligations of others. Joint and several liability is therefore more common when there are few and closely connected borrowers, such as married couples entering a mortgage contract. Japanese joint municipal bonds are also issued under joint and several liability, in this case because there is a very strong fiscal redistribution system in Japan that in effect ties the fiscal strengths of the prefectures closely together. The German joint *Länder* issues, meanwhile, have several but not joint liability.

In a ruling of September 2006 ([15]) the German constitutional court set extremely high hurdles for mutual fiscal support among German states. That ruling drew substantially on the idea of moral hazard to come to the conclusion that federal fiscal redistribution measures are not suited to the remedy of debt sustainability problems. In essence, the German court ruled that local fiscal autonomy is difficult to reconcile with joint fiscal liability. The Japanese local finance system, which was reformed after the Yubari shock of 2007, foresees an increasing level of oversight and control of local authorities when their fiscal situation deteriorates. As a result, local fiscal flexibility is diminished in situations where joint fiscal responsibility may become more necessary. The German argument would therefore not contradict the approach taken by Japan, but just reflects different choices about the limits of local authority.

At the EU level, the issue of joint debt issuance has been discussed multiple times. The European Community (the predecessor of the European Union) reacted to the economic aftermath of the 1970s oil crisis by issuing bonds to refinance loans to energy, industry and infrastructure sectors (Decision 78/870/EEC). Known as the Ortolí facility⁶, the loans were made to companies across the European community. The bonds were liabilities of the EC, not of the member states, and their repayment was backed in the first instance by the repayment of the loans that were made with the funds raised. The European Community budget provided additional credit support to these bonds with the result that they carried AAA credit ratings. Two aspects differentiate these bonds from the German joint *Länder* and Japanese joint municipal bonds, however. First, the issuing entity is not a consortium but a single entity. Unlike the members of the issuance consortia, the EC members have no explicit obligation to provide the funds to repay the bonds. Instead, the EC member states have an unspecific obligation to provide the EC with funds for the budget, something known as a sufficiency guarantee. Second, the EC bonds were not fiscal instruments used to directly finance expenditure. As a result, there was no need for a fiscal capacity to repay them, such as a tax or tariff arrangement. The bond liability was designed to self-liquidate through the receipt of loan repayments. This model of raising funds for loan provision was so successful that the European Union is still issuing bonds under this model. Now as then, these bonds are obligation of the EU as a single legal entity, not of the member states.

In the aftermath of the 2008 financial crisis, efforts were made to design a new form of common borrowing instrument in the EU, this time for fiscal expenditure purposes, and with some form of joint liability of member states. No model was found to be both workable from a market perspective and compliant with the legal constraints

⁶Strictly speaking, the programme was called the 'New Community Instrument' in the ever romantic Brussels vernacular.

of the European Union Treaty framework. The seminal document on this process is the Green Paper on ‘stability bonds’ [24] which was produced with large-scale involvement of private sector experts. It concluded that joint and several liability structures conflict with the no-bailout clause (Article 125) of the Treaty on the Functioning of the European Union (TFEU), so a Treaty change would be required to implement them. Other models under discussion at the time, such as tranching of government bonds, are unworkable in practice⁷.

One way in which to reconcile the prohibition of joint and several liabilities in Article 125 TFEU⁸ with the principle of fiscal assistance of Article 122⁹ is that triggering mutual guarantees arising from joint and several liabilities amounts to an ex-post fiscal transfer. A member state that wishes to obtain fiscal support from others can receive such support ex-ante through the provisions of Article 122. If joint and several liabilities were available, the same member state could obtain the same money through the issuance of such a debt instrument, and subsequently refuse to repay it. This would trigger corresponding payments from the other borrowers, amounting to ex-post fiscal support. The approach taken by the EU framework is simply that because ex-ante support is available, ex-post support adds nothing as far as mutual support is concerned. At the same time, it creates uncertain future obligations on national budgets that should be avoided.

The extent to which debt in the EU can be mutualised is also a constitutional law question related to the notion of sovereignty for each of the member states. The right to raise taxes, and to decide how they are spent, is a fundamental aspect of sovereignty. The putative motto of the Boston Tea Party, ‘No taxation without representation’, is an example of that. The EU framework does not provide for a full-scale transfer of national sovereignty to the EU, and fiscal authority is not included in the competencies transferred to the EU. The expenditure of the EU is therefore always pre-approved by the member states. In a 2011 ruling related to the EFSF [16], the German constitutional court opined that not only a transfer of fiscal sovereignty to the EU had not taken place, but that the German parliament was barred from enacting such a transfer by the German Basic Law. This is just one example where, even if EU law were not an obstacle to joint and several liability bonds, national constitutions may stand in the way.

16.3.2 Supranationals

While direct joint debt issuance by multiple sovereigns appears difficult, there are several examples of multi-national entities that can issue debt. In many cases, such debt

⁷The problems with tranching of government bonds are discussed in more details in Section 24.2.5 on page 259.

⁸‘A Member State shall not be liable for or assume the commitments of central governments, regional, local or other public authorities, other bodies governed by public law, or public undertakings of another Member State, without prejudice to mutual financial guarantees for the joint execution of a specific project.’

⁹‘Where a Member State is in difficulties or is seriously threatened with severe difficulties caused by natural disasters or exceptional occurrences beyond its control, the Council, on a proposal from the Commission, may grant, under certain conditions, Union financial assistance to the Member State concerned.’

issuance is backed by commitments of the nations that constituted the entity (labelled here as members). Examples are the World Bank Group (issuing debt out of one of its subsidiaries, the International Bank for Reconstruction and Development, IBRD), the European Investment Bank (EIB), the European Stability Mechanism (ESM) and the European Bank for Reconstruction and Development (EBRD). This segment of issuers is known as the supranationals, or supras. Together with regional and agency borrowers, the bonds issued by supranationals form a bond market segment known as SSA (supras, sub-sovereigns and agencies). The SSA segment is characterised by medium-sized issuance programmes (smaller than most government markets), a reasonable level of secondary market liquidity, and high credit quality compared to private sector issuance. In this sense, the SSA segment forms a bridge between the largest and most liquid government markets and high-grade private sector debt.

Not all supranational institutions issue debt, or at least not debt that is publicly tradeable. Two well-known examples are the International Monetary Fund (IMF) and the Bank for International Settlements (BIS). In both cases, the independence of operations from the vagaries of private debt markets is at least part of the *raison d'être* of these institutions.

A common feature of a supranational is that it does not have a fiscal function and instead raises funds for on-lending. While this on-lending may have concessionary features compared to direct market borrowing for the recipients of such loans, the loan terms still generally cover the cost of the supranational's funding and other operations. In principle, therefore, a supranational should not be dependent on ongoing support of its members. As a result, commitments of members to the supranationals are not quantifiable contingent liabilities of those members and so do not affect their debt levels directly.

Member support for supranationals usually takes the form of paid-up and callable capital. Paid-up capital is, as it would be for a private bank, invested at least partially in highly liquid assets. This liquidity reserve is designed to ensure timely payment of outstanding bonds without an immediate need to refinance in the bond market. Lending operations are funded by debt issuance. Should the available capital levels fall below requirements, members may be asked to provide additional capital. As a result, the members do not need to guarantee the debt issued by the supranational to ensure high ratings, instead they commit to replenish the capital base of the issuer when needed. Such capital calls should be the exception, not the norm, but depending on the supranational in question, they can have other political motives. If members decide to forgive part of the loans made by the supranational, they have to offset the value lost through new capital injections. In this way, nations can fund debt relief efforts through the injection of new capital into multilateral lending institutions.

16.4 GOVERNING LAW AND CONTRACTUAL CLAUSES

Bonds can be issued in the country where the borrower is resident, using the laws of that particular country to settle potential disputes between the borrower and the bond investors. Such bonds are called domestic bonds. Sometimes, however, it may be more advantageous for the borrower to issue bonds in a different market, or investors

prefer to lend money on the terms of a well-known legal system. This has been done for centuries¹⁰, but the market for such off-shore issues got an important boost from capital controls imposed by the US government and in particular the introduction of the Interest Equalization Tax (IET) in 1963. This led many US issuers to issue bonds in the London market and the resulting large market in foreign bond issuance is now called the eurobond market or the euromarket. This is now somewhat confusing because there is also the euro currency and euro bond can mean a euro-denominated issue issued in whatever market or a eurobond issue denominated in whatever currency. When precision is required, one talks about bonds as being euro-dollar, euro-yen, euro-sterling, and so on. This convention is also the reason why interest rate futures on short rates have 'euro' in their names, such as euro-dollar and euro-yen contracts. This means that the short rates refer to off-shore deposits, i.e., deposits in the euromarket. This does not necessarily mean that there are interest rates differences between the on-shore and off-shore markets.

There are informal names for off-shore bond issues issued in a particular country in this country's currency. Such names are yankee (US), samurai (Japan), kangaroo (Australia), bulldog (UK), etc. Conversely, a bond issued by a US entity in the euro market is called a reverse yankee. A particular type of samurai aimed at the Japanese retail market is called uridashi.

Bond investors usually buy bonds with a view to holding them for a long time. This means that they are exposed to changes in the business outlook of the issuer as well as to the conditions of the legal system in which the bond is issued. In order to reduce the risk of a dramatic negative change in the likelihood that the bond will pay as promised, bonds usually carry covenants that restrict the freedom of the issuer to carry out certain actions such as asset disposals, dissolution, and so on. Violating a covenant usually means that the bonds become immediately due and payable and other penalties may apply. The conditions applying to a particular bond are documented in the prospectus of a bond which is circulated to investors at the time a bond is sold. The prospectus usually also contains background and financial information about the issuer that can help making an informed investment choice. The exact requirements for a prospectus depend on the market in which a bond is sold or listed.

Typically, a so-called preliminary prospectus is distributed during the marketing period for a new bond, i.e., before the actual prospectus is distributed. The preliminary prospectus usually has a warning printed in red on its front page, stating that the prospectus is not final. For this reason, preliminary prospectuses are also called reds in the market. Frequent issuers do not produce independent prospectuses for each bond issue. Instead, they establish an issuance programme with a common prospectus that has placeholders for issues of varying conditions. Individual issues under such a programme then simply refer to the programme prospectus and add individual terms and conditions (T&Cs in short) to each issue. Where prospectuses have to be registered with

¹⁰For instance, the first Japanese long-term government bond was a sterling denominated bond issued on 23 April 1870 in order to raise finances for the first railways in Japan. The bond had a volume of 1 million pounds and a coupon of 9 percent. It was collateralised with customs revenues and three railway lines.

supervisor authorities, the use of such omnibus prospectuses is also known as shelf registration.

Sovereign issuers generally do not publish prospectuses for domestic law bonds although they may publish short terms and conditions for individual issues. This is consistent because the general conditions applicable to domestic debts are generally regulated by the relevant civil law and financing law for the government. Investors need to be aware, however, that sovereigns can change the laws governing domestic debts. One important driver of the importance of the eurobond market is that investors can pursue defaulting issuers in a jurisdiction that is perceived as independent.

Some clauses keep reappearing in bond prospectuses, so one should be aware of them. Here is a selection of the most important.

Governing Law The law by which legal disputes between the bond holders and the issuer are governed (note that this may be different from the laws both the country where the bond is issued and the country where the issuer is domiciled). English Law or New York State Law is most commonly used for international bonds for historical and practical reasons¹¹.

Jurisdiction The courts that will have jurisdiction over any disputes. This is somewhat different from Governing Law and usually mainly related to the cost of legal representation. Potential plaintiffs generally prefer courts close to their home location because retaining legal representations in other locations may incur additional cost and uncertainties.

Events of default A list of events that will be considered breaches of the contract between the issuer and the investors. Such events will provide reason to pursue the issuer in court. The most important is of course failure to pay, but other possibilities can include breaches of covenants.

Cross-default Statement that defaults on one debt issue are deemed to be events of default on all or all similar debt instruments. This protects bond investors because it means that they are immediately able to enjoin in bankruptcy proceedings even if the next payment under the specific instrument they hold that could be missed is still a long time away.

Collective action clause In the event that the issuer needs to restructure a bond, there are essentially two ways in which to implement such a restructuring. The traditional way is for the issuer to make an offer to bondholders (usually after some discussion with them and their advisers) and force them to seek redress in court if they deem the offer unacceptable. The downside of this approach is that, depending on jurisdiction, individual bond holders can cause substantial problems for the issuer through series of court cases that can delay a successful restructuring. Recently, therefore, bond prospectuses contain clauses that set

¹¹One has to be aware that different countries can have very different laws and that some legal concepts may not exist in all countries. English law is well known, generally not subject to abrupt changes, and English law courts operate in other countries. English law also recognises a floating charge, i.e. a charge on an unspecified asset. Most jurisdictions only recognise a fixed charge on a specified set of assets.

out rules for collective decisions by bond holders which are then binding on all of them.

Rating triggers Some corporate bonds contain clauses that lead to higher coupon payments in case of credit rating downgrades. Such clauses are designed to protect investors from valuation losses caused by the spread widening that usually accompanies rating downgrades, especially if such downgrades lead to ejection from certain bond market indices. It should be noted that these clauses effectively subordinate the holders of bonds by the same issuer that do not carry these clauses because a larger share of the issuer's cash flow will be diverted away from them after a downgrade.

Negative pledge Promise of the issuer not to pledge any assets as security for future debt issues. This is very important because the assets of the issuer are all that bond investors can look to in case the issuer stops paying on the bond. Negative pledge clauses usually carve out mortgages so that issuers can in the future purchase real estate assets with loans secured by these assets. Such exceptions highlight the substantial impact a negative pledge clause can have on the business activities of an issuer.

Soft bullet When issuers are essentially solvent but cannot manage the redemption payment on an outstanding bond at the time of maturity, they can document the bond in such a way as to permit the redemption at a later date. This is essentially a conditional extension of the bond maturity and may require higher coupon payments during the extension period. Soft bullet structures are essentially a form of credit rating arbitrage. By obtaining an option to pay principal later, the issuer is more likely to make full payment which can support a given credit rating. The downside for investors is that they hold a paper that will give them longer exposure to the issuer precisely at the time when the economic situation of that issuer has deteriorated. In the covered bond market, soft bullets are increasingly being replaced with conditional pass-throughs, discussed in the section on covered bonds (page 272).

Gross-up and tax call This clause used to be standard in Eurobonds. All bond coupons are paid without deducting taxes, but if the issuer is suddenly forced to withhold taxes, the issuer has the choice to either increase the coupon so the net amount paid stays the same (gross-up) or call the bond immediately. Because the European Union has decided to introduce withholding tax, it is no longer practical to have this sort of clause and nowadays bond prospectuses generally specify no gross-up as outlined below.

No gross-up The issuer generally rules out extra payments to compensate for withholding tax.

Indemnity Statement that the law firm drawing up the prospectus and the investment bank distributing it are not liable for actions of the issuer. This does not mean that the investment bank has no responsibility for the prospectus, but it means that an investor trying to sue the bank usually will have to prove that the bank made a deliberate mistake or did not exercise due care and diligence.

A prospectus usually has to be filed with, or even approved, by some regulator before bonds can be sold to the general public. The reds typically are distributed before such approval has been granted, but the content normally is the same as the final version.

16.5 BOND MARKETS

Bonds are traded in public or private markets. In order to trade bonds, it must be possible to agree on the price for a given amount of debt. The most common form of expressing a price is as a percentage of the remaining principal of the bond that is being traded (bonds can amortise and the remaining principal may be different from the borrowed amount). The remaining principal is also referred to as the par amount and prices are therefore usually expressed as percentage of par. A price of 100, i.e., 100% of par, is also referred to as par (e.g. 'this bond trades at par').

Many bonds specify that traded par amounts have to be an integer multiple of a fixed amount. In the past, this was due to bonds being traded in paper certificates of a certain denomination. With the advent of electronic certificates, this technical restriction is no longer valid and most liquid market can trade in any size down to the lowest relevant currency unit (one cent or penny, etc.). Where minimum denominations still play a role is in bonds that are marketed at institutional investors only. By choosing a large minimum denomination (for example, EUR 100,000), the issuer can make it clear that the instrument should be out of reach for retail investors.

Price quotations usually (but not always) exclude accrued interest. Figure 16.1 on page 138 shows that accrued interest is essentially a straight-line interpolation of interest payable between two coupon dates, something that will be discussed in more detail later. The quoted price excluding accrued interest is called the clean price, clean price plus accrued interest is called the dirty price. Dirty price times par amount gives the invoice amount that must be paid in exchange for the given par amount of bonds. Note that every market has rules for the rounding of accrued interest.

While for analytical purposes the dirty price is more useful than the clean price, the clean price has a nicer dynamic relative to yield levels (yields are discussed in more detail later). Figure 16.3 shows that a bond that trades at the same yield as its coupon rate has a clean price close to par while the dirty price shows a saw-tooth pattern. At yields above the coupon rate, the clean price will be below par while the dirty price may be above par. Conversely, at yields below the coupon rate, the bond will trade above par. The yield that makes the clean price equal to par is therefore called the par yield.

Although the correspondence between coupon rate and par yield is close, it is not an identity. Jumping ahead a little bit, Figure 16.4 shows the clean prices of two bonds as a function of time where each bond trades always at a compound yield equal to its coupon. Because accrued interest is calculated in a linear way while yields are used in a power, there is a marginal divergence of the clean price from par in between coupon dates. The difference is larger at higher coupons, and the chart also shows that it is larger in leap years when there is a longer time between coupons.

Market participants tend to agree that it is undesirable to quote bond prices at an arbitrary precision. As a consequence, there is a minimum price increment. In most

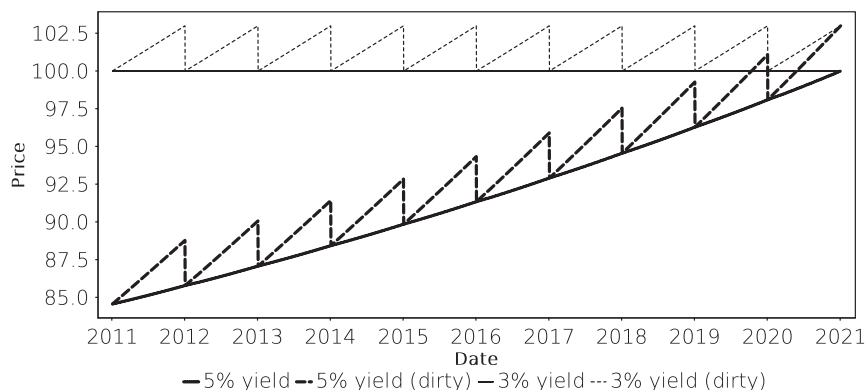


FIGURE 16.3 Clean and dirty prices of the same bond (3% annual coupon) at two different yield levels through time.

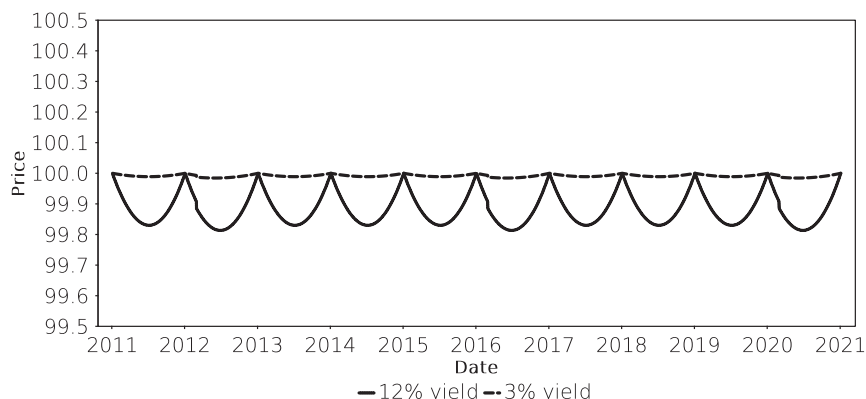


FIGURE 16.4 Clean prices of two bonds (3% and 12% annual coupons) trading at a yield equal to their coupon rate as a function of time.

bond markets this is one tenth of a cent (0.001% of par). The notable exception to this convention is the US bond market which has so far resisted decimalisation and trades in a convention known as thirty-seconds¹². In the US market, a price might be given as 99-24 (note the dash instead of the decimal point) which in decimal terms is $99 + 24/32 = 99.75$. The modern US system of thirty-seconds has evolved, however, because price increments of $1/32$ of one percent of par (around 3 cents in decimalised markets) are too large given the liquidity of the US Treasury market. The first innovation is the plus which is an extra $1/64$ th (around 1.5 cents in decimal

¹²Trading in binary rather than decimal fractions dates back to the days when coins were made out of precious metal and could be split when required. It is simply easier to cut a coin into four parts than into ten. This is also the reason why bond coupons traditionally were binary fractions.

terms) on top of the price in thirty-seconds. In the example above, 99-24+ would be $99 + 24/32 + 1/64 = 99.765625$ ¹³. For even finer price quotations, thirty-seconds can be augmented by eighths of a thirty-second. These are simply appended to the thirty-seconds quote, so 99-245 would be $99 + 24/32 + 5/256 = 99.76953125$. Note that 99-244 is the same as 99-24+ and neither 99-248 or 99-249 exist (99-248 would be 99-25).

Some CBOT Treasury bond futures options use a different convention where the third digit is the rounded decimal fraction in quarters. Here, the 5 in 99-245 would stand for 0.5 of a thirty-second, i.e. 99.245 in the option would be 99.244 or 99-24+ in bond price terms and 99.247 ($99 + 24.75/32$) would be written 99.246 if it was a bond price. Somewhat luckily, 99-242 is the same in either convention.

The Japanese government bond market is an example of a market that trades in yield terms. The minimum price increment is one tenth of a sen (0.001% of par), but in practice yields are quoted in half basis point steps which provides a slightly more coarse price grid.

16.5.1 The primary market

The primary market is the market where bonds are issued and sold to the investor public for the first time. When bonds are sold or bought by investors after they have been placed for the first time, this is referred to as secondary market trading, discussed below.

Bonds are for the most part fairly exchangeable and investors therefore can usually choose whether to buy a new issue in the primary market or a similar bond in the secondary market. This means that issuers need some way to attract investor attention at those times when they wish to place new supply. The foremost incentive is price. New issues are usually sold at a yield that is slightly above the fair value implied by surrounding issues from the same issuer. This yield pick-up, the new issue premium, is an important measure of investor demand. When issuers offer high new issue premia, it can be a sign that investors have concerns about the issuer. New issue premia also affect secondary market levels. When they are sufficiently high, it may be attractive for investors to participate in the primary supply only to immediately sell the new bond in the secondary market¹⁴. This will then put pressure on the new and old bonds alike.

Investors usually have an interest in large issues because they are traded more actively, leading to more efficient price formation. To achieve this, issuers in many cases sell new bonds in multiple tranches. A new bond is first sold as an initial tranche and then subsequent tap issues are sold that are fungible with the initial tranche. Fungible means that the various tranches are fully exchangeable with each other and fungibility must be specified by the issuer. There are no requirements for each tranche of a bond to be sold in the same way. For instance, it is quite common for governments to syndicate the first tranche of a new bond and then auction subsequent ones. Some issuers offer additional bond tranches on a continuous basis to a group of dealers. The issuer will

¹³There is no 'minus' because, for instance, '99-24-' can be expressed as 99-23+.

¹⁴This practice of flipping also exists in other markets.

then typically give the dealers a pricing grid for a selection of bonds that are available through reverse inquiry.

For corporate bond issuers and smaller sovereigns, the way to sell bonds is to hire investment banks who will market and sell the new bond through a process called syndication. In a syndicated deal, one or more banks are appointed as lead manager and these banks generally work with other banks as sub-distributors. The issuer pays the banks a fee that is a small percentage of the cash raised. Fees depend on the maturity of the deal and the credit risk involved. Depending on issue size, issuers will usually attempt to put together a syndicate group that has global distribution capabilities. This would normally mean picking a domestic bank in the issuer country with a strong domestic distribution network and one or two global investment banks one of which might, for instance, be stronger in Europe and the US and the other stronger in Asia. The hierarchy of investment banks in a given deal is generally is split in lead manager, co-lead and syndicate member. The level of responsibility does not only coincide with the fee distribution, but is also an important marketing factor. Some research firms prepare league tables that show the total number of deals a bank has been involved in, as well as the total deal volume. Such league tables generally place more weight on lead mandates than, for instance, co-leads. Another important reason to prefer lead manager status is that new bonds are sometimes oversubscribed, i.e., there is more investor demand than actual bond supply. After all the investor demand indications have been collected, an allocation process starts where each investor is assigned part of the new issue. The issuer and lead managers can choose to prefer certain clients in the allocation process and although this generally does not have the same economic implication as IPO allocations in the equity world, it is important for banks' client relationships that they are seen as being able to satisfy their client's wishes.

Price formation in the syndicated market can be done in two ways. In a bought deal (also known as a firm underwriting commitment), the lead banks guarantee a price or spread level to the issuer. If the banks are unable to distribute the issue to investors at this price level, the banks purchase the bonds for their own account. This places a high degree of risk on the banks and this is generally reflected in higher underwriting fees. In the alternative best efforts approach, investment banks solicit client interest but do not guarantee placement.

Initially, the banks will decide together with the issuer on the issue size and a so-called spread guidance, i.e., an indicative spread range relative to a well-known benchmark. For instance, spread guidance may be that x bn of bonds will be sold at a spread of 23–25 bp over benchmark B . The investment banks will then solicit bids from investors who may indicate that they would purchase y_1 million at 24 and y_2 at 25 bp spread. This is known as the book-building process. The investor bids are aggregated and if, for this example, there are at least x bn bids at a level of 24 bp over the benchmark, the issue will be priced at this level for all investors. Spread guidance and even the size of the issue can be adjusted during the book-building period which can last from less than a day to a few weeks. Increasing issue sizes during the book-building process is disliked by some investors, and issuers and syndicate banks sometimes rule out this option by marketing a deal as 'no-grow'.

Once sufficient client interest has been collected, the new issue books are closed and the issue is priced by observing the benchmark price and adding the relevant

spread. The pricing of a new deal is usually seen as the decisive moment in a new bond deal. The issue price so found is known as the fixed-price reoffer price because the syndicate banks usually commit to buy back the new bonds at this price for a limited time period known as stabilisation. The price effectively remitted to the issuer is the all-in price which is the fixed-price reoffer minus the fees paid to the banks and other costs, such as fees to law firms working on the prospectus or listing fees.

A particular syndication structure called a pot deal has become widespread in Europe after first appearing in the US. In a pot deal, there is one book of investor interest and fees are split according to a fixed formula between the participating banks. This is in contrast to the classic syndication structure where banks do not reveal the names of their clients to the issuer during the book-building. Pot deals put more pressure on the investment banks because the issuer has more information about the progress of the process. At the same time, pot deals avoid situations where one syndicate member tries to 'look good' to the issuer by exaggerating client interest and then ending up buying a large part of the allocation for its own book. When such behaviour occurs, the bank in question will usually attempt to free up its balance sheet by selling the issue in the secondary market after the end of the stabilisation period, which will cause the new issue to underperform and thereby upset investors. Syndicate banks can still participate in pot deal transactions as investors, and the volume absorbed by them is disclosed separately as joint lead interest. Such participation can have various motivations, including the need to have trading inventory available for subsequent market-making in the new issue.

The alternative to syndicated deals is to sell new bonds by public auction. This avenue is only open to large government and sub-sovereign issuers who can rely on enough attention being present when they wish to come to the market. In an auction process, bidders¹⁵ are invited to submit prices (competitive bids) to the issuer or the agent of the issuer who will then allocate the auctioned bonds to the highest bidders. Bidders are usually allowed to submit multiple bids at different prices so their bids can be partially satisfied. In an American auction, each successful bidder pays the price that they bid in the auction while in a Dutch auction¹⁶ each successful bidder pays the same price. This uniform price is determined by the issuer according to predefined rules, e.g. it could be the lowest price at which all bids can be satisfied. Generally, bids at the lowest accepted price are allocated pro-rata among all bids at this price level. Some issuers offer non-competitive bids (bids with no price attached) in conjunction with competitive bids. These non-competitive bids are usually allocated at the average auction price. Confusingly, there is also the concept of non-competitive allocation, discussed below.

The difference between the average price of an auction and the lowest allocated price is called the tail of the auction and in some markets serves as an indication of the strength in demand for the new bond. The longer the tail, the lower the demand because

¹⁵ A minimum standard of financial soundness is required for bidders in any auction. The issuer must have certainty that all bidders can actually pay for their share in the auction, and that bids can be submitted in a reliable and secure fashion.

¹⁶ A Dutch auction must not be confused with a Dutch Direct Auction which is the mechanism used by the Dutch State Treasury Agency to place bonds in the market.

it indicates that a large number of defensive bids had to be filled to place the whole issue. Of course, this sort of logic needs to be adjusted for duration risk (longer maturities should automatically have larger tails) and the tail might be completely meaningless in the presence of large non-competitive bids. Another important indicator for the success of a new issue, in addition to new issue premium and auction tail, is the bid-cover ratio, which is the amount of bids received divided by the amount offered. An auction where less bids were received than the amount on offer, i.e., where the bid-cover ratio is less than one, is said to be uncovered. Syndicated deals are often judged on the mathematically equivalent total book size.

No single indicator is sufficient to judge the success of a syndication or auction. An issuer looking to achieve a high bid-cover ratio, or equivalently, a large book, can simply offer a higher new issue premium to entice participation. This risks poor performance of other bonds after the supply, and thereby the alienation of long-term investors. Conversely, the issuer may want to keep the new issue premium low and so potentially face a low bid-cover ratio or even uncovered issuance. Syndicate members individually or collectively may wish to make a deal more successful than it would otherwise be and place larger bids for themselves. This pushes up the bid-cover ratio at a given new issue premium.

Most auctions are conducted in a way that makes them more similar to syndication. The issuers appoint bidding groups of banks and only members of the bidding groups are allowed to submit bids. In some cases, such as Germany, membership in the bidding group is effectively open to all banks, so there is only a marginal relationship between the issuer and the banks. In other cases, the bidding group consists of so-called primary dealers which are banks that have been conferred a special status by the issuer. When there is a primary dealer group, the issuer will usually expect a certain minimum of bonds being taken down by each primary dealer; otherwise the primary dealer status will be withdrawn. The primary dealers are thereby incentivised to place the new supply just as a syndicate member would be. However, primary dealers are only rarely paid fees for auction participation and instead in some cases buy bonds at auctions over the market level at the time. Compensation by the issuer instead happens through syndicated deals, other trading mandates, etc.

Some issuers use non-competitive allocations to compensate successful bidders. In this case, there is a certain point in time after the auction where successful bidders can request additional bonds at the price paid at the auction in proportion to the volume they successfully received at the auction itself, e.g., 10% of that amount. So, successful bidders are given a call option. Generally, the various incentives given to bidding group participants mean that although there is no publicly disclosed payment of syndication fees, the economic relationship can be very similar.

The German government bond supply system stands out from other large issuers such as the US, Japan, or France for the absence of a formal primary dealer group. The requirements to become a member of the 'bidding group' at the German Debt Management Agency are minimal. One reason is that the Agency is itself active in the secondary market for its bonds. It can retain bonds at auction and has therefore an additional parameter to control new issue pricing in addition to the cut-off price and the share of non-competitive bids it allocates. Faced with weak bidding for new bonds, it can set the cut-off price near the top of the bidding range and allocate a high share of

non-competitive bids or retain a large share of the new supply. Conversely, it can allocate more competitive bids through a lower cut-off prices and lower non-competitive allocation. In effect, the Agency can use its own balance sheet in a way that is very similar to a dealer syndicate in a bought deal.

When a regular issuer, in particular a government, has a number of bonds outstanding, the most recent issue is called the on-the-run issue and the preceding one the off-the-run. It is also common to talk about the first off-the-run, second off-the-run, and so on, and to attach maturities, e.g. on-the-run 10 year, etc. In some markets, notably the US, Italy, Spain and, to a lesser extent, France, the higher liquidity of on-the-run securities is associated with a liquidity premium (cf. Chapter 18).

16.5.2 The secondary market I: (interdealer market)

Trading in bonds is generally not done on public exchanges although most public bonds are officially listed to satisfy investor constraints or central bank collateral eligibility rules (for instance in the case of the ECB). Some exchanges further require that off-exchange trades are reported to the exchange which will then publish information about those trades. However, trading is generally done by bilateral agreement between two parties, or through broking.

When securities houses deal among themselves, they are sometimes reluctant to do so directly in order to avoid giving away information about their current positioning. This is the reason for the existence of brokers (also called inter-dealer brokers) who try to match buying and selling interest between dealers anonymously. Trades through a broker can involve name giveup where both sides are informed after the trade is agreed as to who the counterparty is, or be completely anonymous with each dealer facing the broker as their counterparty. Brokers are usually paid a small fraction of the traded volume in fees.

Bond brokerages have over the last few years built electronic trading systems that increase the efficiency of the brokerage process. On electronic trading platforms, dealers can display prices anonymously and deal on the prices displayed. In some cases, government issuers have encouraged banks to set up electronic market making platforms such as MTS. Such platforms combine an electronic trading system with commitments to display firm prices on the platform for a certain amount of time per day so as to provide liquidity in the interdealer market. Although the commitments are usually less strict than what would normally be market convention, the presence of such trading systems means that smaller securities firms who would have difficulties risk managing a full trading operation can use a certain minimum level of interdealer liquidity as a backstop when offering liquidity to their customers. This is the reason why government issuers are so interested in supporting the existence of such platforms, while large securities houses face the dilemma that by supporting the interdealer market, they are effectively supporting their competition in dealing with customers. In practice, the volumes traded on such platforms are a fraction of total market volume.

By and large, interdealer platforms bring more advantages to all of their participants than disadvantages. A large global investment bank has stress-tested this assumption in 2004 by executing a series of trades in a market making platform called EuroMTS for euro area government bonds that added up to over EUR 15bn bought within a few

seconds and then selling part of this amount back to the market a few minutes later. Because the market makers were required to show certain minimum size quotes across a large number of bonds, this bank was able to execute this large volume without giving the other banks the opportunity to adjust the prices to reflect this large demand. In particular, the trade was aimed at executing more volume than was available at that price in the usual hedge instrument for euro area government bonds in this maturity sector, the Eurex Bund futures contract. The initiating bank therefore was probably able to make a sizeable profit from the liquidity mismatch of EuroMTS versus the futures exchange. Consequently the trade was probably the main impetus for tighter market regulation.

However, the important lesson from the episode is that trading on EuroMTS is still continuing. Although some rules have been introduced to limit the scope for such trades reoccurring, dealers seem to value the presence of the platform enough not to withdraw.

16.5.3 The secondary market II: (customer-facing market)

Customers deal with securities houses in a variety of ways. Voice trading is still common, but generally used only when large transaction sizes are executed. Normally clients do not talk to traders directly, but through the intermediation of a salesperson. This is has a few practical as well as compliance-related reasons.

The bulk of customer trading these days is done through electronic platforms because this is vastly more cost-efficient and less error-prone than voice trading. An electronic trading system can automatically feed trade information through to the backoffice systems managing the settlement instructions and the risk management systems of the bank. Among electronic platforms, one distinguishes between quote request and streaming systems, as well as between single-dealer and multiple-dealer platforms. In a quote request system, a client can solicit price quotes from a number of dealers, but then has to arrange execution of the winning quote in other ways. In contrast, a streaming trading system is a system where firm price quotes are displayed on an ongoing basis and a customer can trade on these quotes through the same system.

16.6 ACCRUED INTEREST

Bond coupons are generally paid to the current holder of the bond in one sum. When bonds are traded between coupon dates, this means that although the seller has effectively been lending money to the issuer, the interest since the last coupon payments will be paid in full to the buyer. In trading, the seller therefore generally is compensated for the amount of interest that has accrued since the last coupon payment date.

Bonds generally pay interest only on good business days, but unlike swaps, bonds usually do not adjust the coupon for the actual accrual period.

Accrued interest is calculated exactly like interest on a CD, using the coupon rate C as the interest rate. Hence, accrued interest is simply:

$$I = C * DCF(t_{i-1}, t_s) \quad (16.5)$$

where t_{i-1} is the last coupon payment date and t_s is the settlement date of the bond transaction. The DCF here is calculated according to the rules for the respective bond market, which may be (and usually is) different from the convention for the money market. The actual/actual convention is quite common in the bond market, so if there are coupon periods of over one year, the fictitious accrual period outlined in the money market section applies. A good example are German 10Y government bonds which usually have a first coupon period of more than one year.

When a bond is traded only in multiples of a fixed notional amount, the bond has a long or short first or last coupon period and the issuer, in the prospectus, commits to pay a rounded amount on the first or last coupon corresponding to the fixed notional, Equation (16.5) is incorrect for the respective coupon period. Accrued interest is instead calculated on the actual (rounded) coupon payment using the appropriate daycount fraction.

Some bond markets (notably the UK and in some sense Japan) still use ex-dividend periods. These stem from a time when coupons were physical pieces of paper that had to be cut off and mailed to the issuer to receive payment. The ex-dividend period is the time just before the coupon payment date when the coupons are assumed to have been cut off and travel in the mail. If a bond changes hands during the ex-dividend period, on the payment date, the original holder will get the coupon, not the buyer of the bond. For settlement during the ex-dividend period, the accrued interest is negative because the buyer will hold the bond for a few days without getting paid any interest. The formula Equation (16.5) effectively becomes:

$$I = C * DCF(t_i, t_s) \quad (16.6)$$

(Note the change in the index of the coupon date). In Japan, negative accrued interest does not exist because although an ex-dividend period exists, no bond settlement takes place during that period.

Bonds in most markets are quoted on a clean price basis. This means that for the purposes of settling a trade, one has to calculate accrued interest and add it to the price quoted in the market before multiplying with the notional amount. The sum of clean price and accrued interest is called dirty price. Schematically,

$$\text{Dirty price} = \text{Clean price} + \text{Accrued Interest} \quad (16.7)$$

16.7 YIELD

The general idea of a yield is that it should measure the rate of return on a bond investment. Multiple ways of expressing such returns exist and they differ in the underlying simplifications they make. One can follow the evolution of yield measures by adding complexity to the mathematics. The dirty price of a bond in percent of notional will be denoted by P and the coupon C . t_n is the maturity date as used before. Yields are denoted by y .

16.7.1 Running yield

The simplest yield calculation is:

$$y = \frac{C}{P} \quad (16.8)$$

The idea here is that if one pays 101% to receive a coupon of 4% running, one has a rate of return of around $4\%/1.01 \approx 3.96\%$. The running yield is the straightforward equivalent of the dividend yield used by equity analysts.

16.7.2 Simple yield

The difference (or at least one of the differences) between equity and bonds is that bonds usually have a final maturity date. If one pays 101 for a bond today that matures in two years time and therefore pays back 100, the extra 1% of investment is lost unless the coupon makes up for it. The next best yield measure takes this into account by amortising the capital loss over the remaining time to maturity:

$$y = \frac{C - \frac{P-100}{t}}{P} \quad (16.9)$$

This simple yield is the standard yield convention in the Japanese market.

The problem with the simple yield is that it implicitly assigns a capital loss occurring in 10 years time the same value as a capital loss occurring in one year. Ideally, therefore, the capital losses would be weighted by the appropriate discount factor. From this thought, it is just a little step to the idea to weight each cashflow with the appropriate discount factor.

16.7.3 Compound yield

Compound yield is the equivalent of the internal rate of return in corporate finance. Specifically, yield is the solution to the equation:

$$P = \sum_i \frac{Cf_i}{(1+y)^{DCF(t_s, t_i)}} \quad (16.10)$$

A variation used sometimes in Germany is the Moosmüller yield which treats the time period until the first coupon payment as a money market instrument. The price-yield equation in this case becomes:

$$P = \frac{1}{1+y_{DCF}(t_s, t_1)} \sum_{i=1}^n \frac{Cf_i}{(1+y)^{DCF(t_1, t_i)}} \quad (16.11)$$

The Moosmüller yield is not widely used, but has one advantage over the compound yield outlined above. It is market convention (street convention) to treat a bond as a money market instrument during its last coupon period, i.e., when it becomes a pure

discount bond. The Moosmüller yield does this automatically because in this case the sum runs over only one term and that term has a discount factor of 1, leaving the discounting to the initial money market term.

Both yields cannot be calculated in a closed form for arbitrary inputs. In other words, there is no formula that gives the compound or Moosmüller yield for a given price in the general case of an arbitrary number of coupons and settlement date. The only way to calculate yields is therefore to try different values for y in Equation (16.10) or Equation (16.11) to see which works. Of course there are clever ways to do this very quickly, but it is still a lot more expensive computationally than simple yields.

While compound yield appears to be comparatively ‘scientific’, it does in fact have some remaining drawbacks. Equation (16.10) is invariant against changing the timing of cashflows as long as the cashflow is adjusted with the interest rate y . For instance, moving any coupon C of an annual bond forward by one year would not change its present value (dirty price) as long as the coupon is reduced to be $C/(1 + y)$. Moving it back one year would also not change the price as long as the coupon is changed to be $C(1 + y)$. This implies that y is the implied forward interest rate between the two coupon dates¹⁷. In this way one can show that y is not only the internal rate of return of all the cashflows, but also the implied forward rate between any two cashflow days. Actual investment returns will differ because investors are unlikely to be able to reinvest coupons they received at this implied interest rate. Indeed, because long-term yields tend to be higher than short-term yields, an investor who holds a bond to maturity and reinvests all intermediate coupons paid by the bond should not expect to achieve a total return equal to the yield.

The implicit time invariance in the rate of return contained in Equation (16.10) also negates the idea of a yield curve, i.e., the concept of different yields for bonds of different maturities issued by the same issuer. Coupons from different bonds by the same issuer should economically be discounted by the same interest rates, no matter which bond they stem. This concept then leads to the idea of discount factor curves which will be discussed later.

16.7.4 Bond-equivalent yield

US government bonds pay interest semi-annually and annualisation is used to arrive at what is known as the bond-equivalent yield y_B with the equation:

$$y_B = 2(\sqrt{1 + y} - 1) \quad (16.12)$$

Bond-equivalent yields are always lower than compound yields by an amount of $y^2/4$. When comparing the yields of bonds with different coupon periods, it is important to

¹⁷To be precise, the forward rate would be a money market rate which could then technically be quoted in a different daycount convention, depending on the market, and so have a different numerical value.

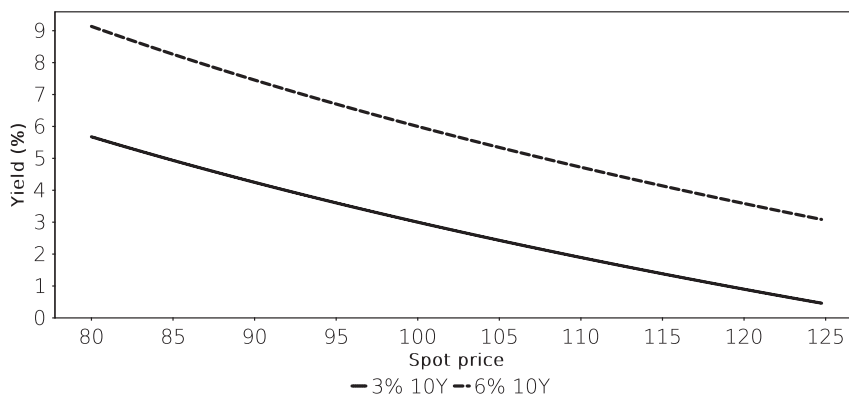


FIGURE 16.5 Price-yield relationship for two 10Y bonds with different coupons.

first translate all yields to the same standard, such as compound or bond-equivalent yield.

In the introduction to this section, it was mentioned that the various yield measures differ in the simplifications they contain, but nothing negative has been said about the compound yield. The problem with any yield-type return measure is the assumption that there is a single interest rate that describes the time value of money. A close inspection of Equation (16.10) shows that any cashflow can be moved forwards or backwards in time as long as the size of that cashflow is adjusted by a discount factor that is always given by the yield y . Inspecting the bond market, however, one finds that bonds of different maturities have very different yields, so the discount factor that needs applied at each point in time is different.

The solution to this problem is to pick one term structure of discount factors that is reasonably realistic, and express differences in returns as spreads to this curve. This is the motivation behind the use of various curve spreads which will be discussed elsewhere.

Yields of fixed coupon bonds generally fall when bond prices increase (cf. Figure 16.5). The shape of this relationship differs between bonds with different coupon rates. Bonds with higher coupons pay more of their total cashflow early in the lifetime of the bond and these cashflows are less sensitive to the discount rate. The bonds with the highest sensitivity to discount rates are therefore zero-coupon bonds. The inverse relationship, i.e., the change in yield corresponding to a given change in price, shown in Figure 16.6, accordingly shows less yield impact of the same price change for lower coupon bonds.

The extreme case of a zero coupon bond best illustrates the trade-off between interest rate sensitivity in absolute and relative terms. As Figure 16.7 shows, the duration (relative price risk) of the zero increases linearly with maturity. However, because the present value of the bond falls, the absolute price risk first increases, but then also falls and eventually converges to zero. Intuitively, an infinite maturity zero coupon bond has no value, no matter what the discount rate is (as long as it is positive).

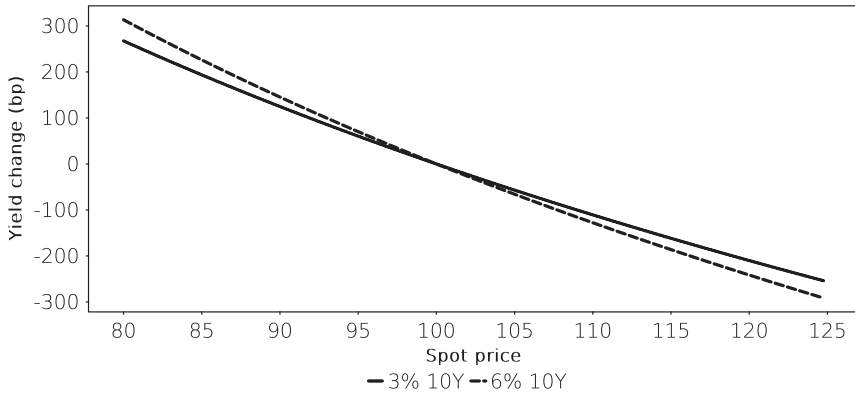


FIGURE 16.6 Yield changes for price changes of two 10Y bonds with different coupons.

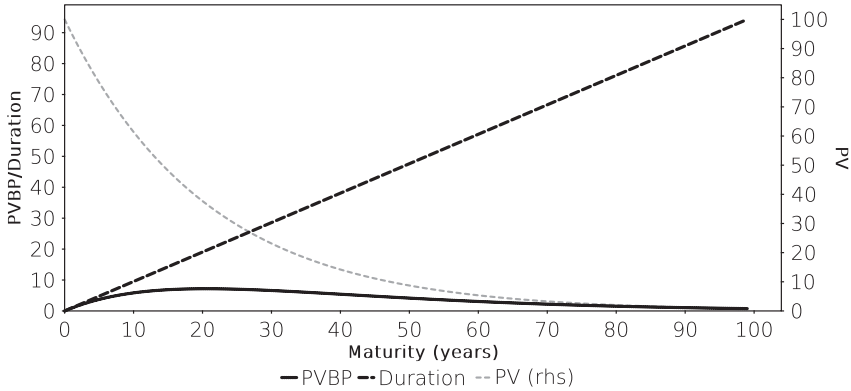


FIGURE 16.7 Price and risk measures for a zero-coupon bond at 5% yield as a function of maturity.

16.8 INTEREST RATE RISK

Equations (16.8) through (16.11) establish mathematical relationships between price and yield. Therefore, changes in yields will drive changes in prices and vice versa. Two sensitivity measures are most common, namely the modified duration D^* defined as:

$$D^* = \frac{1}{P} \frac{\partial P}{\partial y} \quad (16.13)$$

and the PVBP (present value of a basis point), written here as:

$$PVBP = \frac{\partial P}{\partial y} \quad (16.14)$$

Clearly, the only difference between these two measures is the dirty price P . Both measures are used, albeit in slightly different contexts. PVBP is the cash amount by which the market value of a given *notional amount* of bonds changes for a 1 bp change in yields. D^* is the relative price change, but because the notional amount of bonds can be seen as the ratio of market value and dirty price, modified duration is also the cash amount by which the market value of bonds with a given *market value* moves. For a fund manager, it is more important to view portfolio size in terms of the amount of assets under management and modified duration is therefore perhaps a more natural measure. A bond trader monitoring positions, meanwhile, will usually see nominal amounts in the trading book and use PVBP.

16.9 CONVEXITY

Convexity is a common but not very well defined concept in fixed income. It is common because it refers to the idea that the sensitivity of prices to changes in measures such as yields are typically nonlinear. Convexity is the term that describes this nonlinearity. However, there are multiple mathematical definitions of convexity depending on the context. Here, convexity is defined simply as the second derivative of price with respect to yield, i.e.:

$$Conv = \frac{\partial^2 P}{\partial y^2} \quad (16.15)$$

In this way, duration is the slope of the tangent in the price-yield relationship at the current yield, while the convexity is a measure of the curvature of that line at that point.

Convexity is generally seen as beneficial for bond portfolios because for a given amount of interest rate taken, a more convex portfolio provides incremental returns relative to a less convex portfolio when markets are volatile and interest rate is adjusted after yield moves. Market participants know this and factor convexity into their investment decisions (cf. Section 20.2 on page 209).

Although this will be discussed in more detail later, the problem with convexity in this definition is that the concept of yields by construction assumes a flat yield curve.

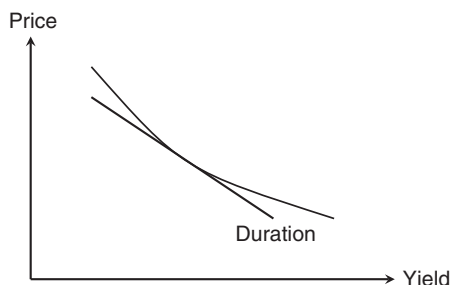


FIGURE 16.8 Graphical representation of convexity.

This means that changes in front end interest rates are in principle weighted equally in the DV01 and convexity risk measures as long-term interest rates. These risk measure simply state risk in terms of so many dollars or euros per basis point in yield change with no regard to relative yield volatilities in different curve sectors. Given the normally higher volatility of shorter interest rates, this means that both DV01 and convexity overstate the interest sensitivity of long-dated bonds.

Generally speaking, convexity is used in the market for all kinds of nonlinear relationships. For instance, the convexity in the convexity adjustment for money market contracts Equation (13.17) has a very different meaning from the one encountered here.

16.10 BOND VALUE DECOMPOSITION

Present value, PVBP and convexity can be better understood by looking at the breakdown of these measures for bonds of different maturities. We assume 5% coupon bonds of 10, 20, 30, and 40 years maturity trading at par and show the breakdown of the present value (the current market price), PVBP and convexity into the contributions provided by principal and coupons. Given the small coupons, the charts sum the coupon contributions per decade (1st–10th coupon, 11th–20th coupon, and so on).

Figure 16.9 shows the breakdown of the bond present value. What may be surprising is that the present value of the first ten coupons is unchanged across all 4 bonds even though they represent all coupons of the 10Y bond but only one quarter of the coupons of the 40Y bond. Mathematically, this should not be a surprise given that the present value of these cashflows is simply ten flows of 5% of notional discounted at a 5% yield for 1 to 10 years. The present value of these cashflows must therefore be a constant.

Somewhat more meaningful is the interpretation of this observation. A 20Y bond can be thought of as a 10Y bond that will be refinanced by the issue of a new 10Y bond in

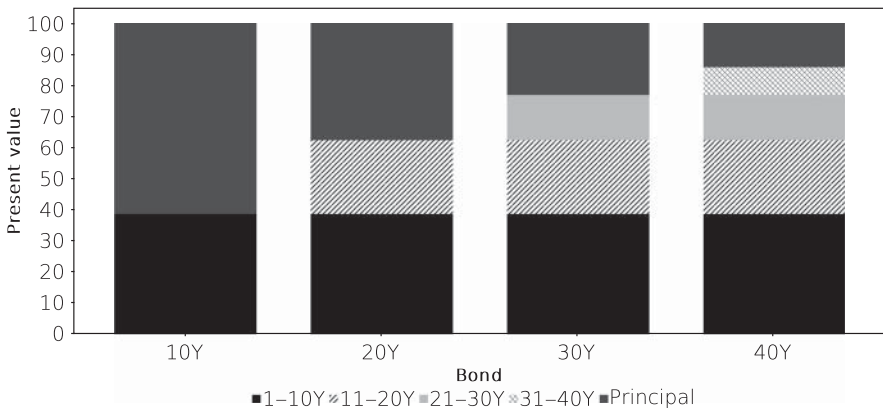


FIGURE 16.9 Breakdown of bond present values by coupons and principals for different bond maturities. All bonds have 5% coupons and trade at par (5% yield).

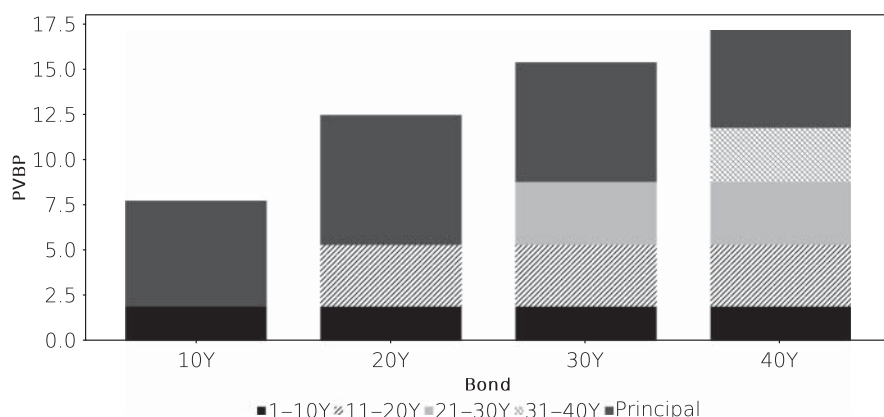


FIGURE 16.10 Breakdown of bond PVBP by coupons and principals for different bond maturities. All bonds have 5% coupons and trade at par (5% yield).

ten years time. The concept of yield as a uniform discount rate therefore implies that an investor in a 20Y bond would be indifferent to investing in a 10Y bond now, and in ten years again in a 10Y provided the expectation that at maturity a bond of equal coupon will again be issued at par.

The breakdown of first-order bond risk (PBVP) in Figure 16.10 is more complex. Longer-dated bonds have more risk and therefore total risk increases with maturity at equal yield. However, the breakdown is not linear. The PVBP of the principal declines with maturity. This is the simple result of the principal itself becoming less valuable when it is being repaid further into the future, something that Figure 16.9 already demonstrated. The PVBP of each coupon block remains constant, again as the simple result of discounting the same cashflows at the same interest rate and then taking a first derivative. In line with the observation on the principal, the PVBP of the coupon blocks declines in the sense that the first ten years of coupons have more PVBP as the next ten years and so on. This again results from the lower present value.

Turning finally to convexity (Figure 16.11), again one observes that the contribution of each coupon block, i.e., the same cashflows at the same time discounted by the same yield is constant. The convexity of longer-dated cashflows is higher, unlike the PVBP. However, the relationship is not monotonous. The convexity of the 40Y principal is lower than that of the 30Y principal. This is a result of the yield calibration used here. Essentially, at very long maturities the decline in present value outweighs the increase in convexity. Taken to an extreme, absent any default risk, an infinite maturity zero coupon bond has zero present value¹⁸ and that present value does not change with the discount rate.

¹⁸The present value of a defaultable principal depends not only on the discount rate but also on expected time to default. This is because principal obligations usually accelerate in defaults.

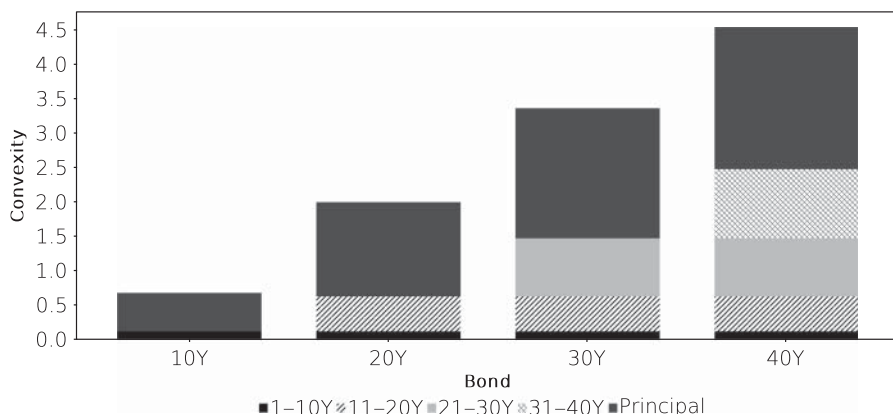


FIGURE 16.11 Breakdown of bond convexity by coupons and principals for different bond maturities. All bonds have 5% coupons and trade at par (5% yield).

16.11 CARRY

Carry is the income earned on a long bond position, taking into account the cost of financing the bond, in other words, coupon income minus financing cost. The usual definition of carry is:

$$\text{Carry} = \text{Spot price} - \text{Forward price} \quad (16.16)$$

Because zero coupon bonds do not accrue any interest but roll up to par, they generally show up as having negative carry in this measure. In some markets, especially the US, one therefore uses yields and:

$$\text{Carry} = \text{Forward yield} - \text{Spot yield} \quad (16.17)$$

Using no arbitrage, the equation to calculate the forward price (under the assumption that no interim coupon is paid) is simply:

$$P_f = \frac{P_0}{Df(t_0, t_f)} \quad (16.18)$$

where $Df(t_0, t_f)$ is given by the repo rate $r(t_0, t_f)$:

$$Df(t_0, t_f) = \frac{1}{1 + r(t_0, t_f)DCF(t_0, t_f)} \quad (16.19)$$

so:

$$P_f = P_0(1 + r(t_0, t_f)DCF(t_0, t_f)) \quad (16.20)$$

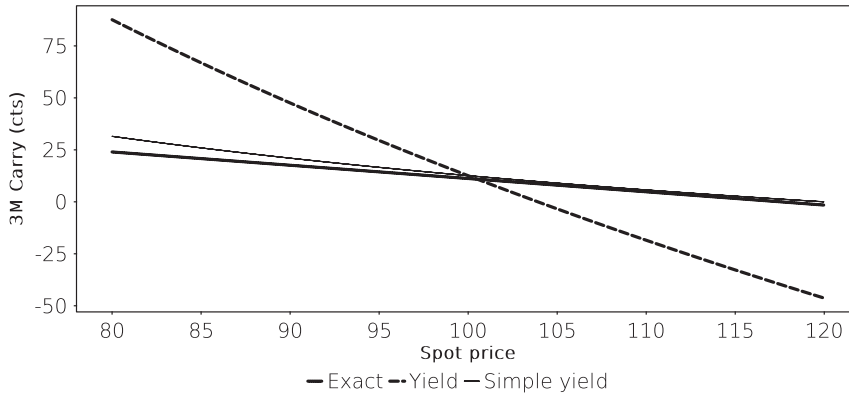


FIGURE 16.12 Carry calculated using the exact method of Equation (16.21) and approximations. Note that using the simple yield gives better approximate results than using the compound yield.

If an intermediate coupon C is paid between t_0 and t_f , there is no immediately obvious forward price because it is not quite clear at which rate the coupon can be reinvested. Standard practice is to use the repo rate $r(t_0, t_f)$ itself, so one has:

$$P_f = \frac{P_0}{Df(t_0, t_f)} - C * DCF(t_{i-1}, t_i) \frac{Df(t_i, t_f)}{Df(t_0, t_f)} \quad (16.21)$$

with:

$$Df(t_i, t_f) = \frac{1}{1 + r(t_0, t_f) DCF(t_i, t_f)} \quad (16.22)$$

Note that Equation (16.21) holds no matter which repo rate is being used. The assumption of reinvestment at the repo rate only enters when the same r is used in all discount factor calculations in Equation (16.21).

The carry calculation using Equation (16.21) is cumbersome and one can instead approximate by multiplying the difference between bond yield and repo rate by the holding period, i.e. de-annualising the relevant interest rates, to arrive at the carry. The quality of this approximation is shown in Figure 16.12.